NTIRE 2026 Joint Denoising and Low Light Image Enhancement Challenge - Track 2 Factsheet- MST++ Based Transformer Network for Joint Denoising and Low-Light Image Enhancement

Varda I Pattanshetty[1,3], Varsha I Pattanshetty[1,3], Nikhil Akalwadi[1,3], Padmashree Desai[1,3], Ramesh Ashok Tabib[2,3], Uma Mudenagudi[2,3]
[1]School of Computer Science and Engineering
[2]Department of Electronics and Communication Engineering
[3]Center for Visual Intelligence (CEVI)
KLE Technological University, Hubballi, Karnataka, INDIA
{01fe22bcs189, 01fe22bcs191, nikhil.akalwadi, padmashri, ramesh_t, uma} @kletech.ac.in

1. Introduction

Joint denoising and low-light image enhancement is an important task in low-level computer vision that aims to recover high-quality images from severely degraded observations. Images captured in low-light environments often suffer from multiple degradations such as strong noise, low visibility, poor contrast, and loss of structural details due to sensor limitations and insufficient illumination.

The NTIRE 2026 Joint Denoising and Low-Light Image Enhancement Challenge focuses on restoring images affected by both noise and illumination degradation. This problem is particularly challenging because the restoration algorithm must simultaneously suppress noise while enhancing image brightness and recovering important structural information.

Recent advances in deep learning have significantly improved the performance of image restoration methods. In particular, transformer-based architectures have shown strong capability in modeling long-range dependencies and capturing global contextual information, which is beneficial for complex image restoration tasks.

In this work, we adopt a transformer-based restoration approach based on the MST++ architecture, implemented within the Retinexformer training framework built on the BasicSR toolbox. The proposed model is designed to effectively remove noise while enhancing low-light images and preserving important structural details. The model is trained using the NTIRE training dataset and evaluated using standard image quality metrics such as PSNR and SSIM.

2. Factsheet Information

2.1. Team Details

- **Team name:** KLETech-CEVI
- **Team leader name:** Nikhil Akalwadi[1,3]
- **Team leader address, phone number, and email:**
  Center for Visual Intelligence,
  KLE Technological University, Hubballi
  Phone: +91 8792188443
  Email: nikhil.akalwadi@kletech.ac.in
- **Team members:**
  - Varda I Pattanshetty[1,3]
  - Varsha I Pattanshetty[1,3]
  - Padmashree Desai[1,3]
  - Ramesh Ashok Tabib[2,3]
  - Uma Mudenagudi[2,3]
- **Team website:** https://cevi.co.in
- **Affiliation with NTIRE 2026 sponsors:** NA
- **CodaBench details:**
  User ID: vkabd
- **Best validation performance:**
  PSNR: 17.168 dB
  SSIM: 0.596
- **Code repository:**
  https://github.com/CeviKle/NTIRE2026-KLETech-CEVI-JointLLIEandDenoising.git

2.2. Method details

2.2.1. General Method Description

Our proposed solution is based on the Multi-stage Transformer (MST++) architecture [1], implemented using the Retinexformer [3] training framework built on the BasicSR toolbox. MST++ is a transformer-based image restoration

model designed to capture both local spatial information and long-range contextual dependencies through attention mechanisms [2].

The network follows a multi-stage progressive restoration framework, where multiple transformer stages are cascaded to gradually refine image features. Each stage consists of encoder–decoder transformer blocks that extract hierarchical representations and improve reconstruction quality. The overall network consists of three main components: shallow feature extraction, deep feature extraction using MST transformer blocks, and image reconstruction.

Given an input degraded image x, the shallow feature extraction layer first maps the input image into a higher-dimensional feature space using a convolution layer:

$$F_0 = H_{SF}(x) \tag{1}$$

where F_0 represents the shallow feature representation and H_{SF} denotes the shallow feature extraction module implemented using a 3×3 convolution layer.

The extracted shallow features are then processed through multiple MST transformer stages that capture global contextual relationships using multi-head self-attention mechanisms:

$$F_d = H_{MST}(F_0) \tag{2}$$

where F_d denotes the deep feature representation and H_{MST} represents the stacked MST transformer blocks responsible for learning hierarchical contextual features.

Finally, the reconstruction module generates the restored image $\hat{y}$:

$$\hat{y} = H_{rec}(F_d) + x \tag{3}$$

where H_{rec} denotes the reconstruction module responsible for producing the final restored image. The residual connection helps preserve low-frequency information and improves training stability.

To further improve perceptual quality and structural consistency, we incorporate Structural Similarity (SSIM) loss together with the pixel-wise L1 reconstruction loss during training. This loss combination helps the model preserve important image structures and textures while effectively removing noise and enhancing low-light regions.

Figure 1 illustrates the overall architecture of the proposed MST++ based joint denoising and low-light enhancement framework. The model takes a degraded image as input, extracts shallow features using convolution layers, processes them through stacked MST transformer stages for deep feature learning, and finally reconstructs the restored output image. denoised output image.

2.2.2. Representative Pipeline

The proposed joint denoising and low-light enhancement pipeline consists of the following stages:

1. Input degraded low-light image containing noise and illumination degradation
2. Shallow feature extraction using a 3×3 convolution layer to project the image into a higher-dimensional feature space
3. Deep feature learning using stacked MST (Multi-Stage Transformer) blocks to capture spatial context and hierarchical representations
4. Multi-scale feature refinement through an encoder–decoder transformer architecture with skip connections
5. Image reconstruction using a convolution layer with residual learning to generate the enhanced image

2.2.3. Training Strategy

The proposed MST++ network is trained using paired low-light and ground-truth images provided in the NTIRE 2026 Joint Denoising and Low-Light Image Enhancement dataset. During training, image patches of size 256×256 are randomly cropped from the training images to form input–target pairs.

The model is optimized using the Adam optimizer with an initial learning rate of 2×10^{-4}. A cosine annealing cyclic learning rate scheduler is adopted to gradually reduce the learning rate during training. The network is trained for 150,000 iterations.

To improve generalization and training stability, data augmentation techniques such as geometric transformations and MixUp augmentation are applied. The training objective combines pixel-wise reconstruction loss and structural similarity loss to encourage both accurate pixel restoration and preservation of structural information.

The final training loss is defined as:

$$L_{total} = L_{L1} + \lambda L_{SSIM}$$

where L_{L1} represents the pixel-wise reconstruction loss and L_{SSIM} represents the structural similarity loss. This loss formulation helps the network effectively remove noise while preserving important image structures during low-light enhancement.

2.2.4. Method Complexity

The proposed method is based on the MST++ architecture, which employs a lightweight multi-stage transformer framework for joint denoising and low-light image enhancement. The network consists of an initial convolutional feature extraction layer followed by multiple stacked MST transformer stages and a final reconstruction layer.

In our implementation, the model uses $n_feat = 30$ feature channels and 3 MST stages. Each stage contains hier-

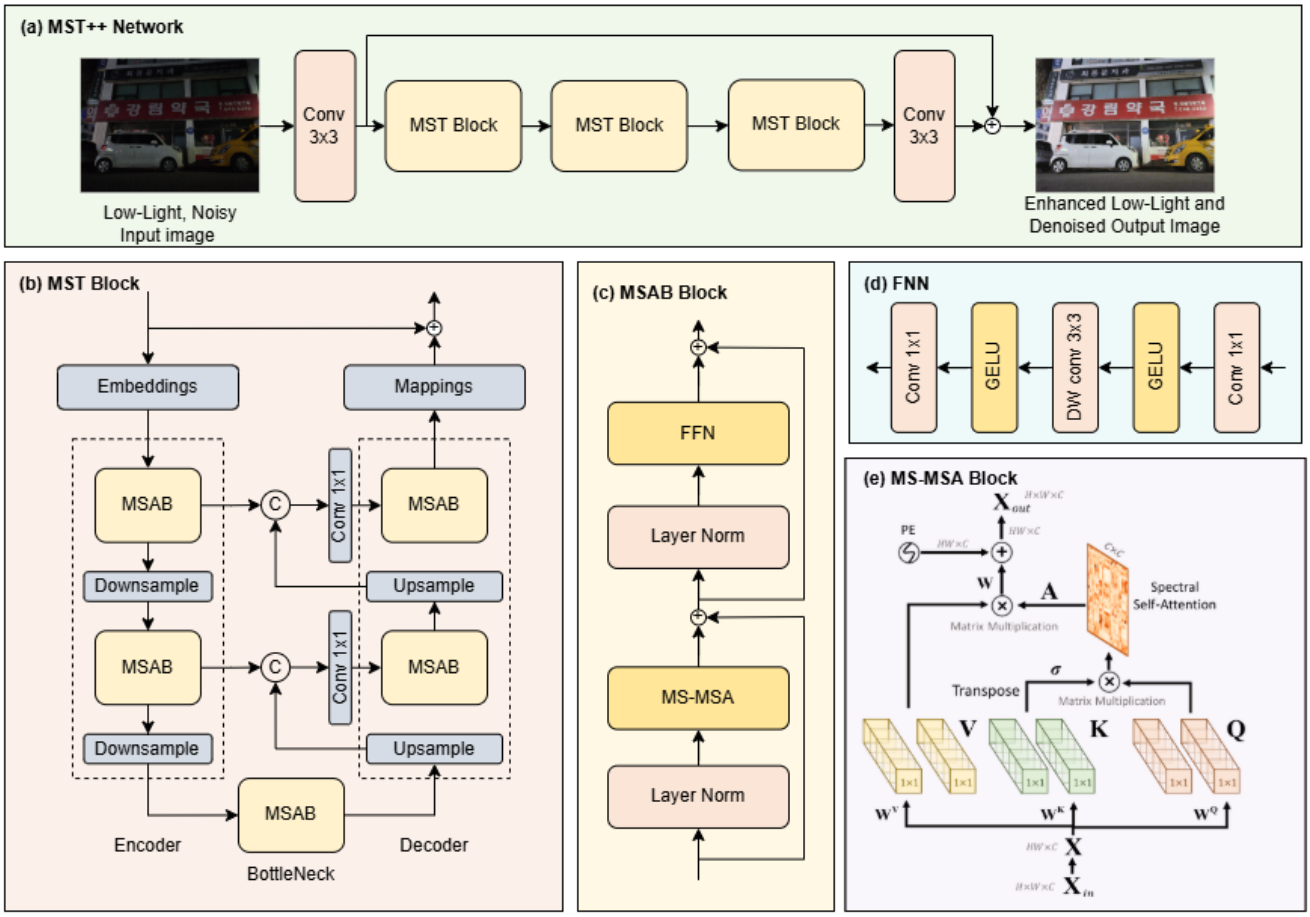

Figure 1. Architecture of the proposed MST++ based network for joint denoising and low-light image enhancement. (a) Overall MST++ network structure consisting of an initial 3×3 convolution layer, multiple stacked MST blocks for hierarchical feature learning, and a final reconstruction layer with residual connection to produce the enhanced output image. (b) Detailed structure of the MST block, which follows an encoder–decoder architecture with downsampling and upsampling operations for multi-scale feature extraction. (c) MSAB (Multi-Stage Attention Block) integrating layer normalization, multi-head self-attention, and feed-forward networks with residual connections. (d) Feed-Forward Network (FFN) composed of 1×1 convolution, GELU activation, and depth-wise convolution for local feature refinement. (e) MS-MSA block illustrating the multi-head self-attention mechanism used to model long-range dependencies in the feature space.

archical encoder–decoder transformer blocks with MSAB (Multi-Stage Attention Blocks) that integrate multi-head self-attention and feed-forward networks for effective feature learning. The hierarchical structure enables efficient multi-scale feature extraction while maintaining moderate computational complexity.

2.2.5. External Models and Additional Data

No external datasets or pre-trained models were used during training. The model was trained exclusively using the dataset provided by the NTIRE 2026 Joint Denoising and Low-Light Image Enhancement challenge.

2.2.6. Training Description

Training was conducted using the PyTorch framework with the BasicSR toolbox. The MST++ network was trained for 150,000 iterations using the Adam optimizer with an initial learning rate of 2×10^{-4}. A cosine annealing cyclic learning rate scheduler was employed to gradually adjust the learning rate during training.

Training patches of size 256×256 were randomly cropped from the training dataset. Data augmentation techniques such as geometric transformations and MixUp augmentation were applied to improve generalization.

The training objective combines pixel-wise L1 reconstruction loss and Structural Similarity (SSIM) loss to encourage both accurate pixel restoration and preservation of

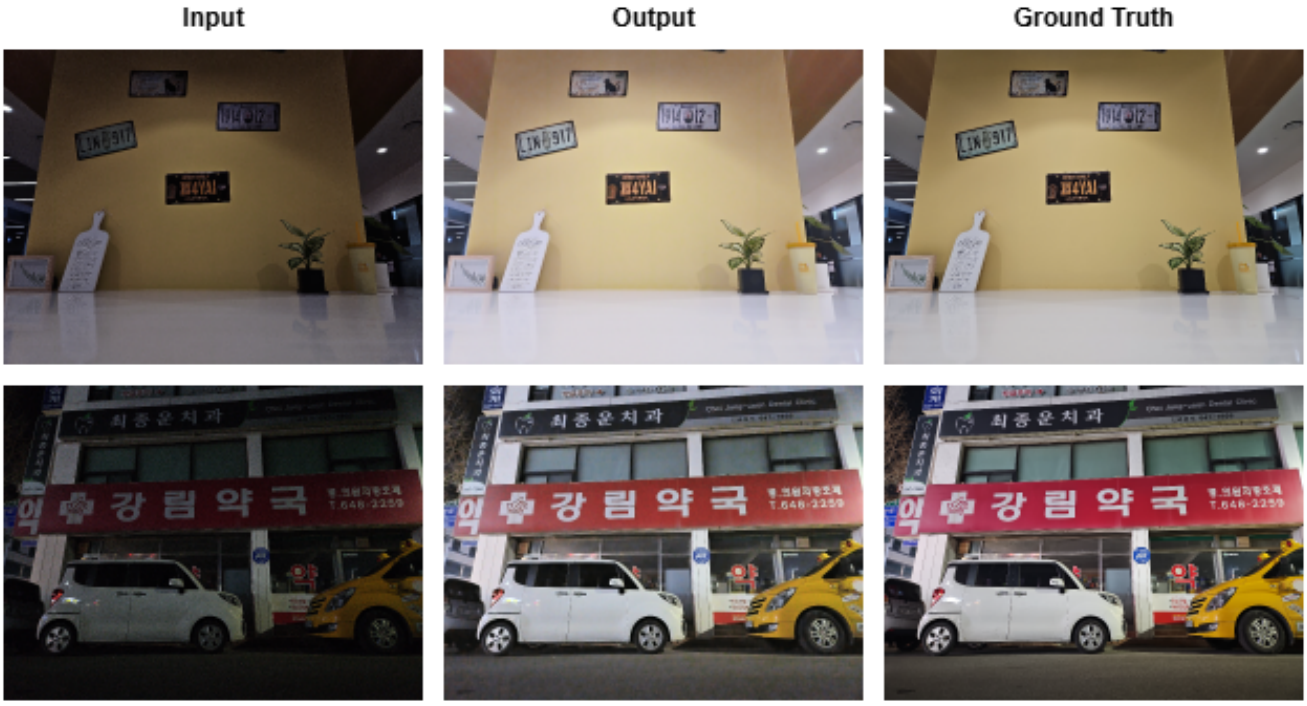

Figure 2. Qualitative results of the proposed MST++ based method for joint denoising and low-light image enhancement.

structural information.

2.2.7. Testing Description

During testing, the trained model checkpoint is loaded and applied to the degraded images from the NTIRE test dataset. Each input image, affected by noise and low illumination, is processed individually by the MST++ network to generate the restored output image.

The network performs joint denoising and low-light enhancement by extracting hierarchical features through transformer-based attention modules. These modules capture both local spatial information and long-range contextual dependencies, enabling effective suppression of noise while enhancing illumination and recovering structural details. Finally, the reconstruction module with residual learning produces a clean and visually enhanced image with improved brightness, contrast, and structural consistency.

2.2.8. Advantages of the Proposed Solution

The proposed MST++ based framework provides several advantages for joint denoising and low-light image enhancement. First, the transformer-based architecture effectively captures both local spatial features and long-range contextual dependencies through attention mechanisms, enabling improved restoration of degraded regions.

Second, the hierarchical encoder–decoder structure allows the network to learn multi-scale representations, which helps in recovering fine image details while simultaneously enhancing illumination in low-light regions.

Third, the use of residual learning facilitates stable training and enables the network to focus on learning the restoration residuals, leading to more accurate reconstruction of clean images.

Finally, the combination of pixel-wise L1 loss and structural similarity (SSIM) loss encourages both accurate pixel reconstruction and preservation of structural information, resulting in visually pleasing enhanced images with reduced noise and improved brightness.

3. Other details

- **Planned submission of a solution(s) description paper at NTIRE 2026 workshop:**
  Yes
- **Other comments: encountered difficulties, fairness of the challenge, proposed subcategories, proposed evaluation method(s), etc:**
  The NTIRE 2026 Joint Denoising and Low-Light Image Enhancement Challenge provided a valuable platform for evaluating modern deep learning-based image restoration methods. The dataset contains challenging low-light scenes affected by both noise and illumination degradation, which makes the restoration task more realistic and difficult.

The challenge setup and evaluation protocol enabled fair benchmarking of different restoration approaches. Although minor difficulties were encountered during model training and testing due to the complexity of jointly handling noise removal and illumination enhancement, the overall organization of the challenge was well structured and facilitated the development and evaluation of effective restoration models.

References

[1] Yuanhao Cai, Jing Lin, Xiaowan Hu, Haoqian Wang, Xin Yuan, Yulun Zhang, Radu Timofte, and Luc Van Gool. Mask-guided spectral-wise transformer for efficient hyperspectral image reconstruction. In *CVPR*, 2022. 1

[2] Yuanhao Cai, Hao Bian, Jing Lin, Haoqian Wang, Radu Timofte, and Yulun Zhang. Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 12504–12513, 2023. 2

[3] Yuanhao Cai, Hao Bian, Jing Lin, Haoqian Wang, Radu Timofte, and Yulun Zhang. Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *ICCV*, 2023. 1